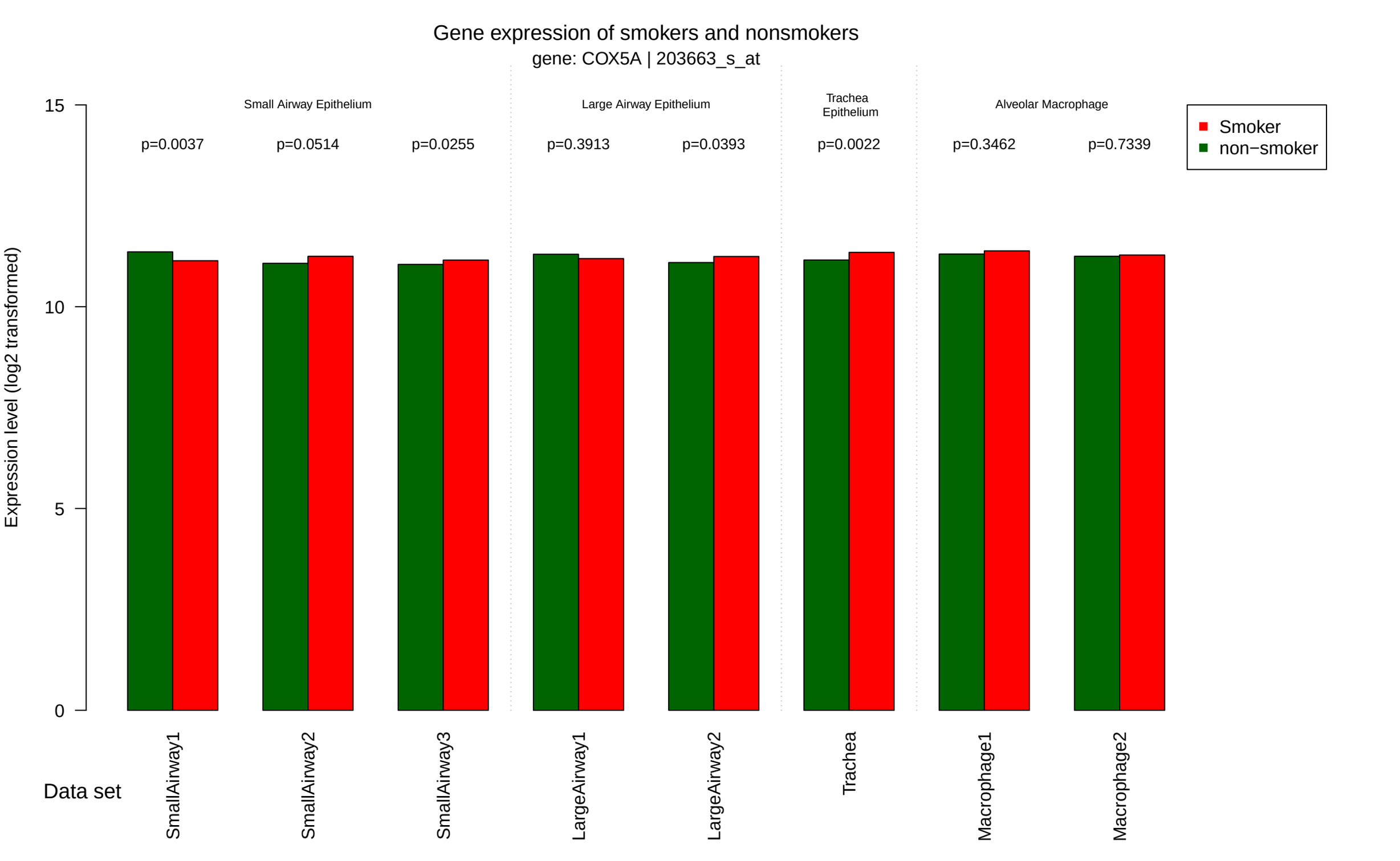
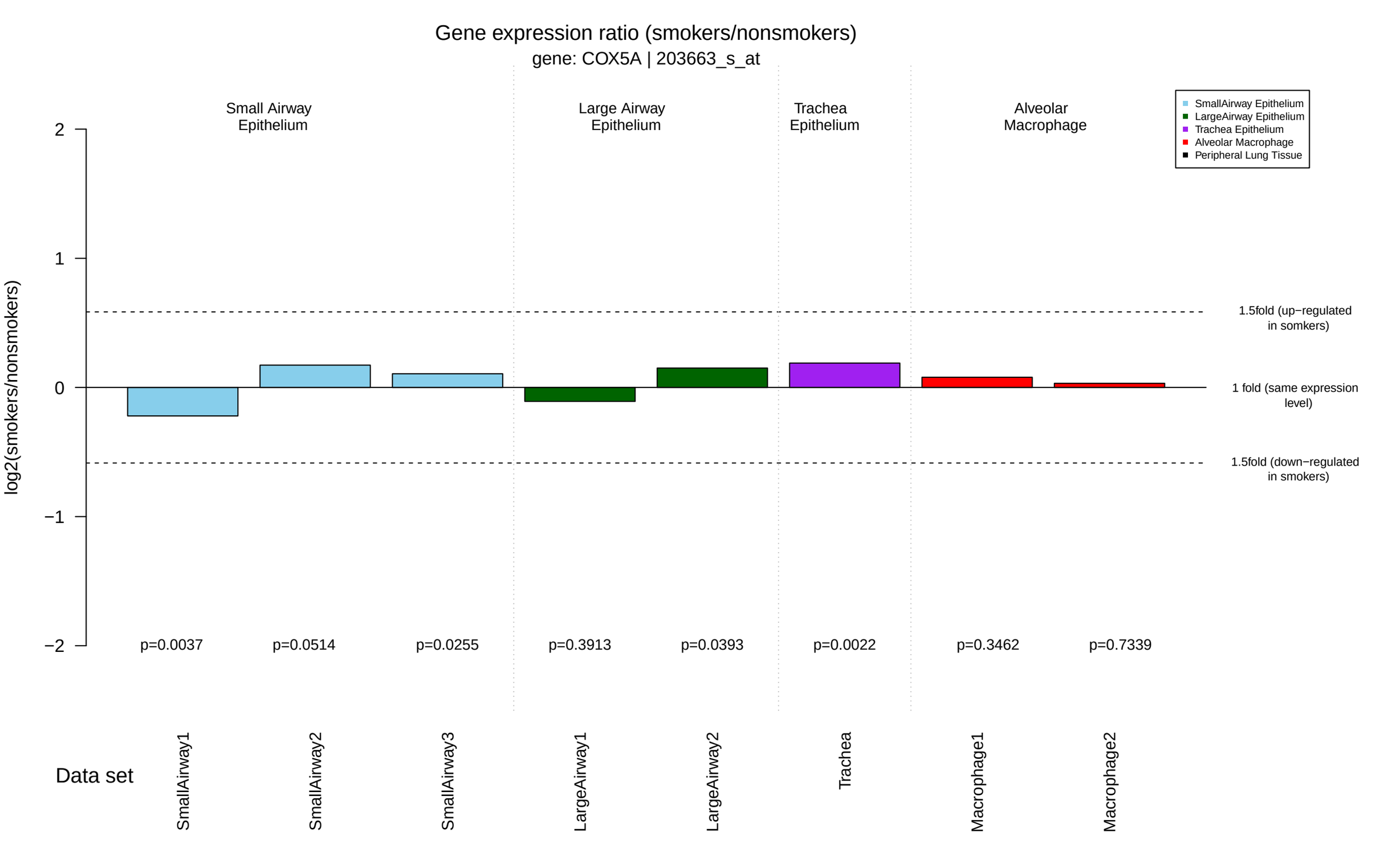
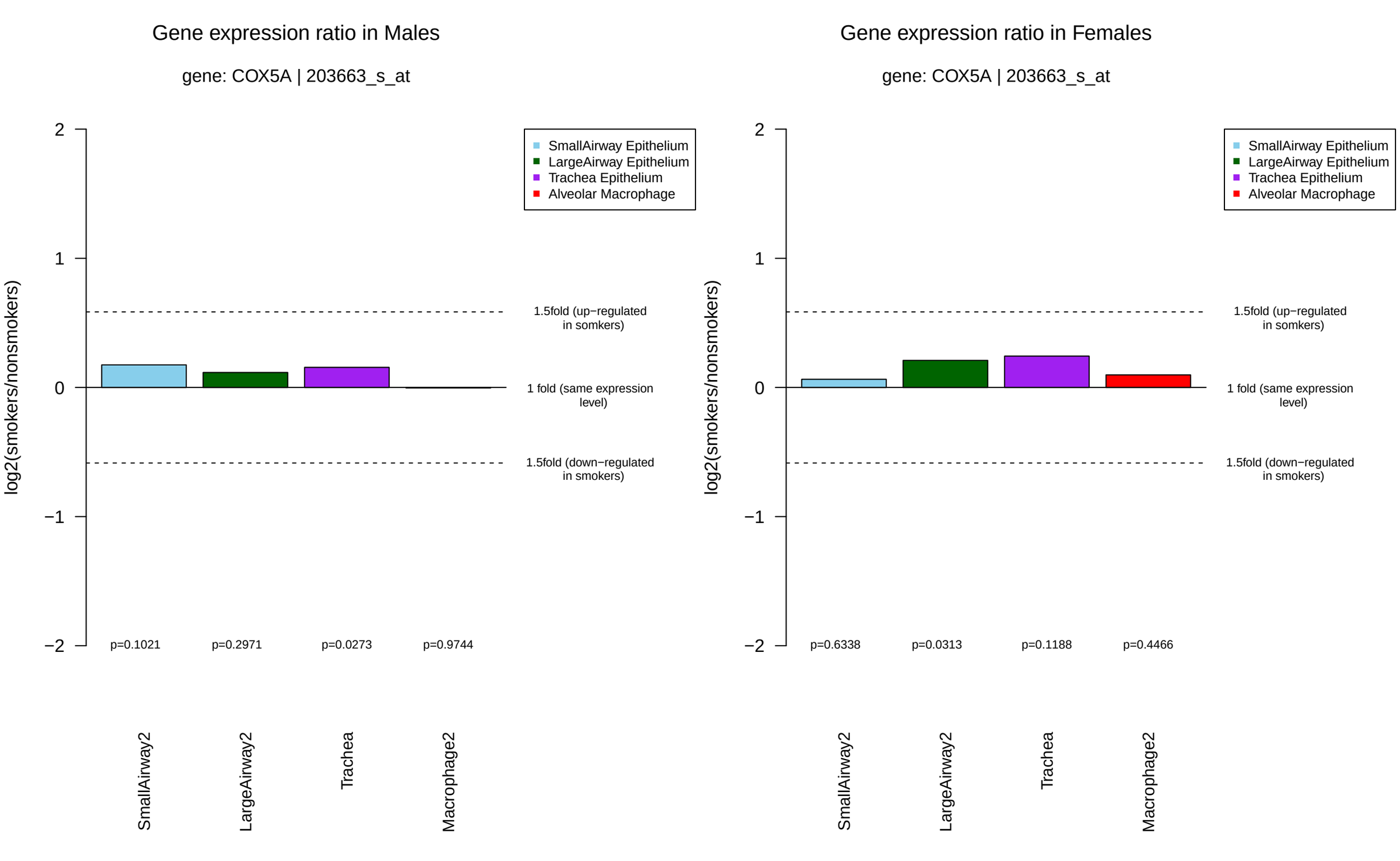
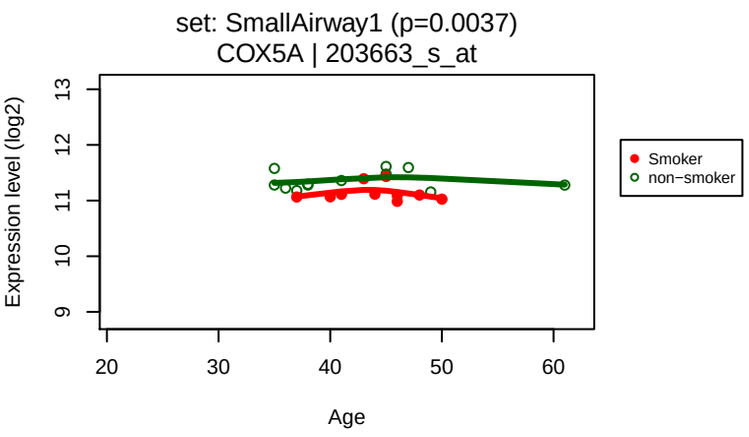
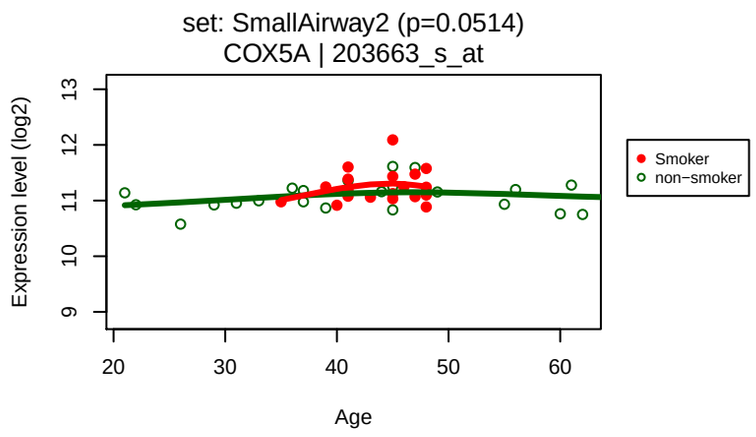
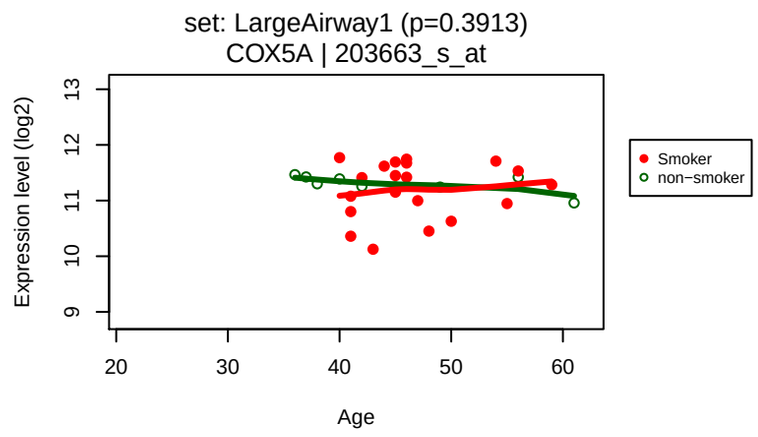

Gene expression of smokers and nonsmokers
gene: COX5A | 203663_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0037
p=0.0514
p=0.0255
p=0.3913
p=0.0393
p=0.0022
p=0.3462
p=0.7339
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

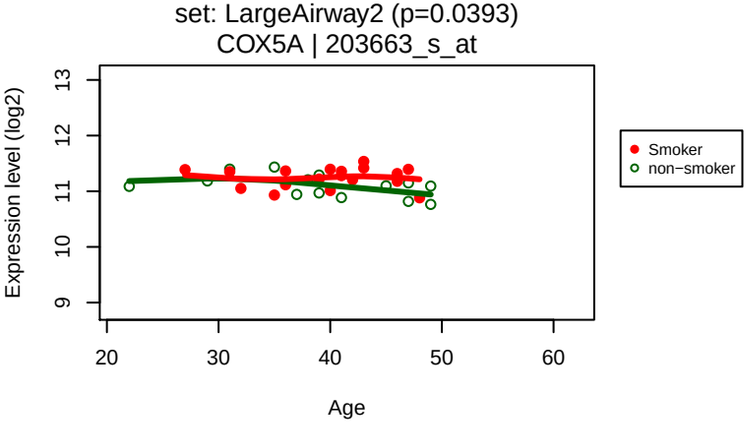

Gene expression ratio (smokers/nonsmokers)
gene: COX5A | 203663_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0037
p=0.0514
p=0.0255
p=0.3913
p=0.0393
p=0.0022
p=0.3462
p=0.7339
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

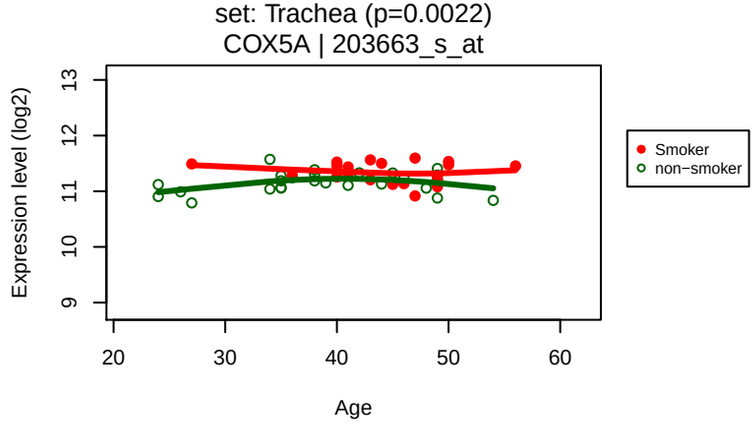

Gene expression ratio in Males
gene: COX5A | 203663_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1021
p=0.2971
p=0.0273
p=0.9744
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX5A | 203663_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6338
p=0.0313
p=0.1188
p=0.4466
SmallAirway2
LargeAirway2
Trachea
Macrophage2

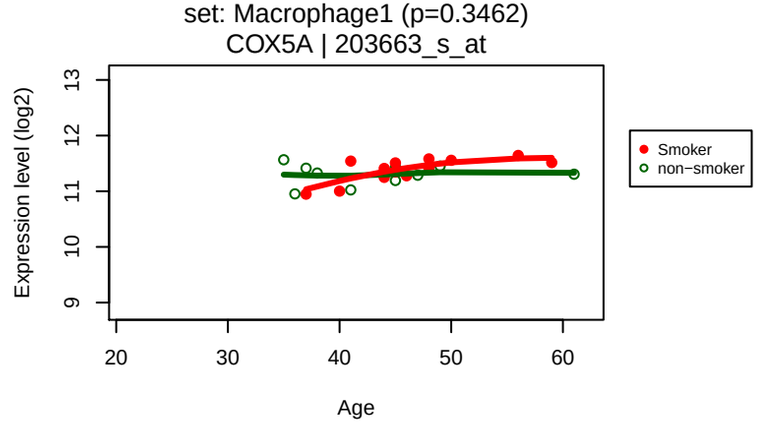

set: SmallAirway1 (p=0.0037)
COX5A | 203663_s_at
Expression level (log2)
Age
Smoker
non−smoker

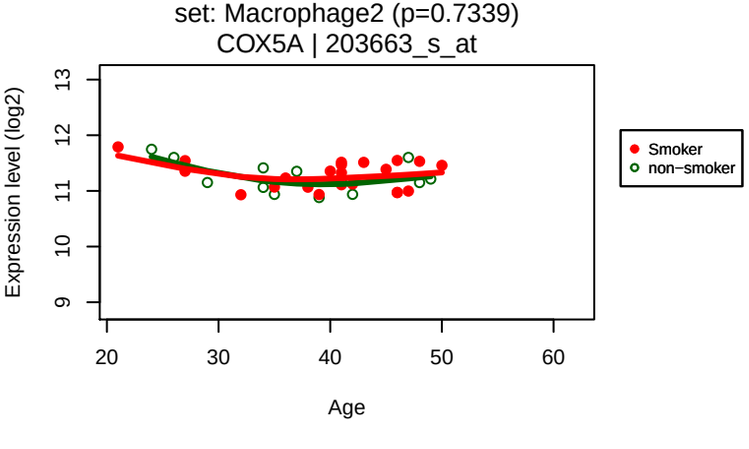

set: SmallAirway2 (p=0.0514)
COX5A | 203663_s_at
Expression level (log2)
Age
Smoker
non−smoker

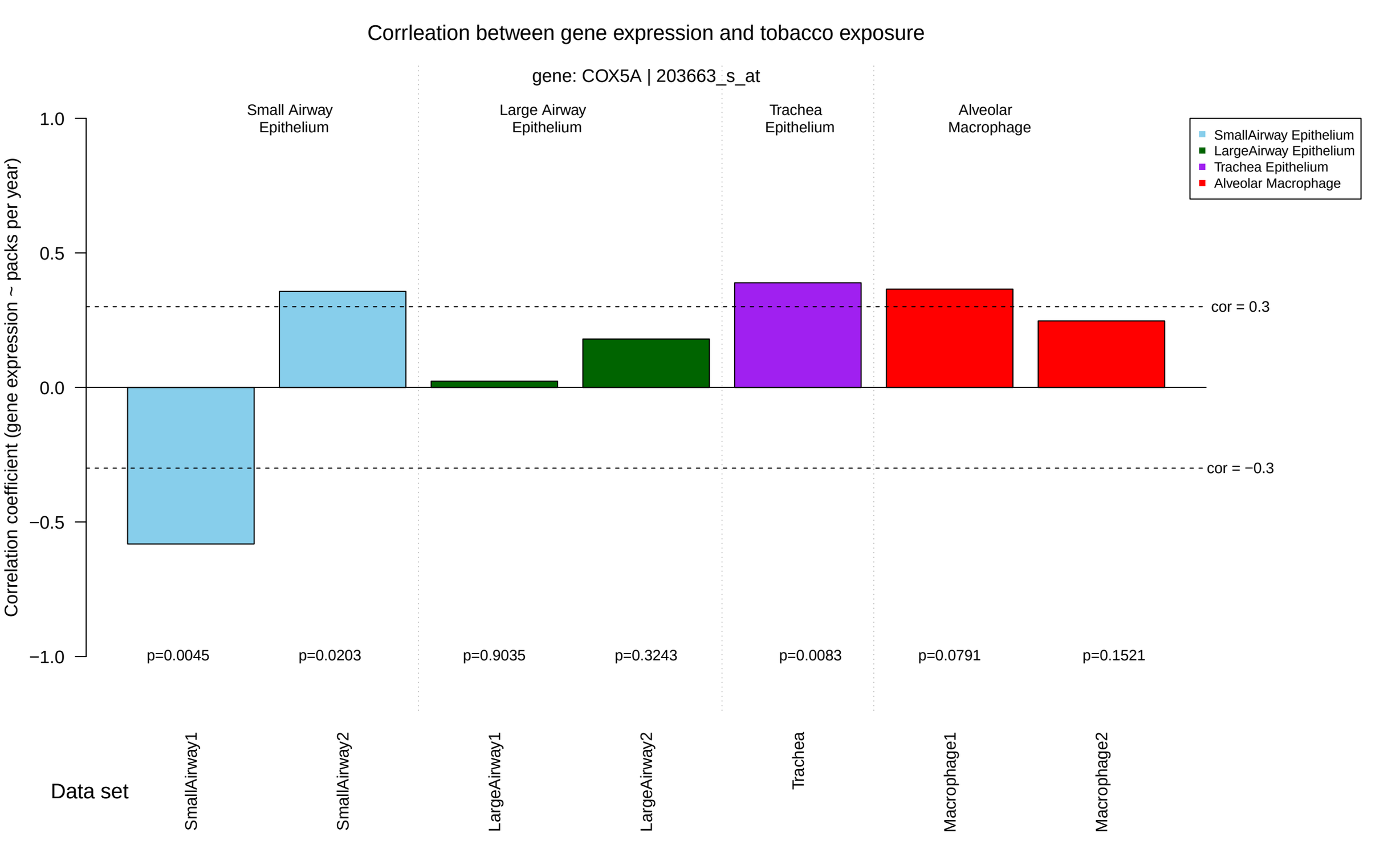

set: LargeAirway1 (p=0.3913)
COX5A | 203663_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0393)
COX5A | 203663_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0022)
COX5A | 203663_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3462)
COX5A | 203663_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7339)
COX5A | 203663_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX5A | 203663_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0045
p=0.0203
p=0.9035
p=0.3243
p=0.0083
p=0.0791
p=0.1521
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2